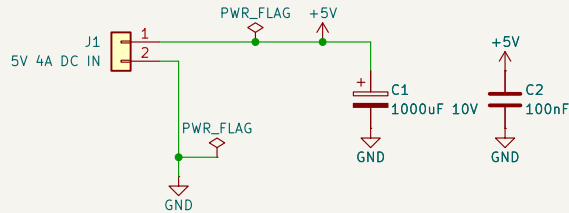
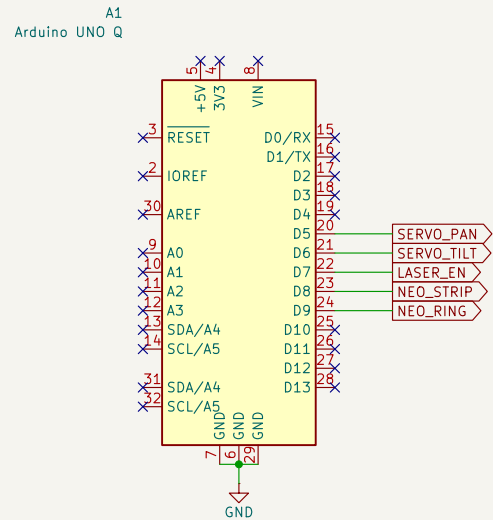


POWER INPUT 5 V / 4 A

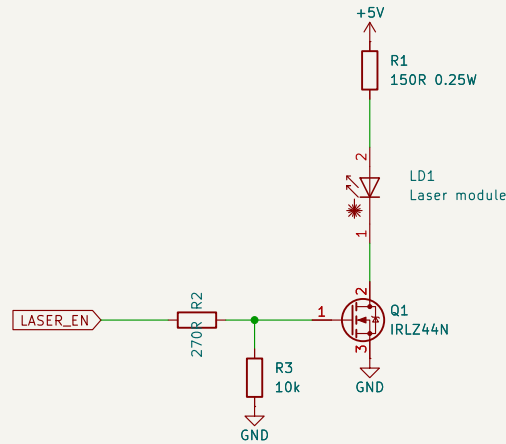


MCU Arduino UNO Q (UNO R3 header layout)

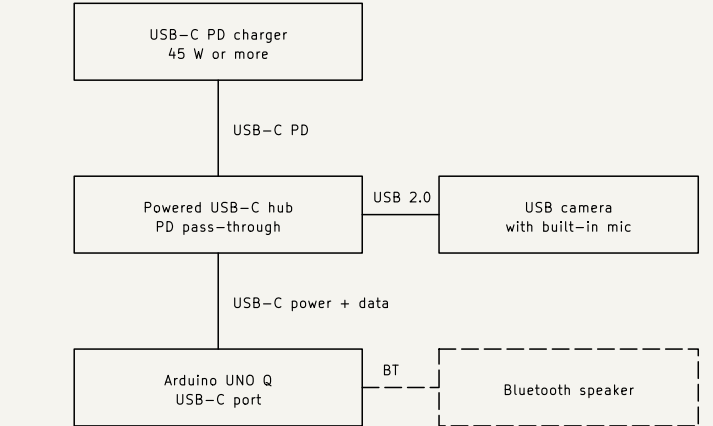


USB-C: power + data from the hub (see SYSTEM)

LASER DRIVER low-side switch



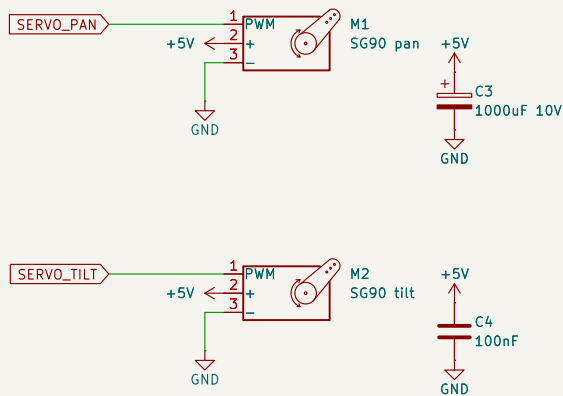
SYSTEM USB and audio



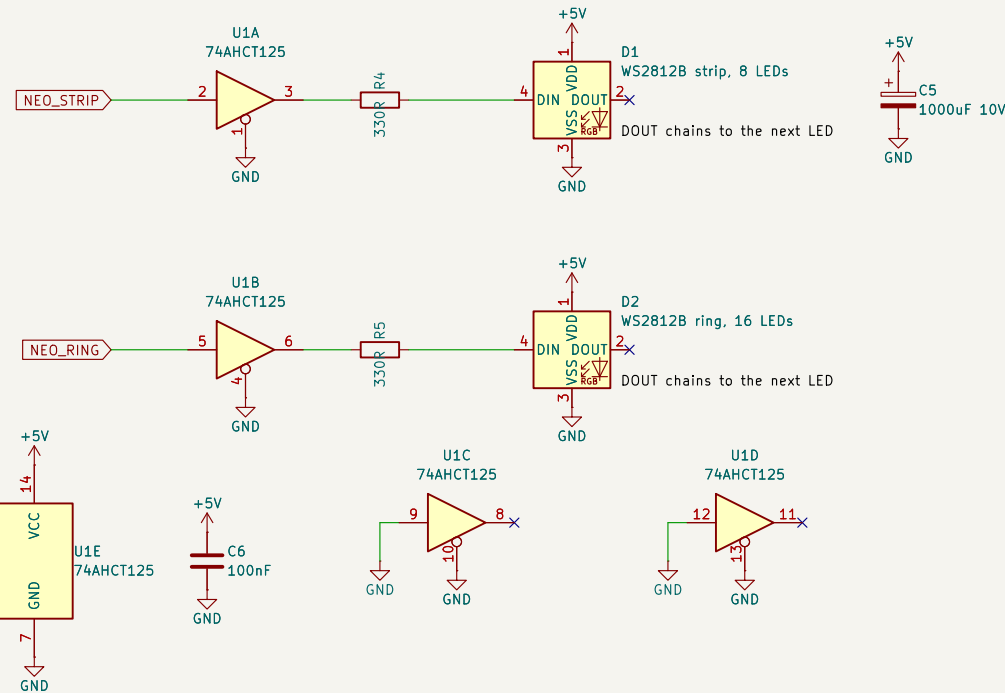
Camera: tool finder video, wake word and speech audio

Speaker: Bluetooth, no wiring

SERVOS pan D5, tilt D6



NEOPIXELS 3.3 V to 5 V level shift



NOTES

- Pin map (matches sketch/sketch.ino): D5 SERVO_PAN (TIM1), D6 SERVO_TILT (TIM3), D7 LASER_EN (PB2, TIMB, plain GPIO), D8 NEO_STRIP, D9 NEO_RING. MCU GPIO is 3.3 V, 5 V tolerant except A0/A1.
- $R1 = (5.0\text{ V} - 2.2\text{ V}) / 20\text{ mA} = 140\text{ R}$, use 150R: about 19 mA, 0.05 W. If the module keeps its own resistor the current is lower; measure before fitting.
- R2: gate charge peak $3.3\text{ V} / 270\text{ R} = 12\text{ mA}$. R3 holds Q1 off during boot and reset.
- LD1 is a salvaged low-power keychain laser module, output power unmeasured. Firmware switches it off after 10 s (sketch.ino and hardware.py). Never aim at people or at head height.
- U1 (74AHCT125, VIH 2.0 V) lifts 3.3 V data to 5 V; WS2812B needs VIH $0.7 \times VDD = 3.5\text{ V}$. Unused gates 3 and 4: inputs and OE to GND, outputs open. R4, R5 at the first LED.
- Servos (about 0.7 A stall each) and LEDs run from J1, never from the UNO Q 5 V pin. J1 GND, UNO Q GND, Q1 source, servo and LED grounds are common.
- Load budget: servos $1.4\text{ A} + \text{LEDs } 24 \times 60\text{ mA} = 1.44\text{ A} + \text{laser } 0.02\text{ A} = 2.9\text{ A}$ worst case.
- All resistors 0.25 W 5 %. Electrolytics 10 V or more.